

## 0.1 Section 1.1 - Intro to Logic

### 0.1.1 Propositions

#### Definition 1 (Proposition)

A **proposition** is a statement that is either

#### Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

**Solution:**

yes

2. Hello.

**Solution:**

No

3.  $x + 3 = 10$

**Solution:**

No, since we don't know  $x$

4. In a right triangle,  $a^2 + b^2 = c^2$

**Solution:**

No, since  $a, b, c$  are not assigned anything

5. In a right triangle having legs of length  $a$  and  $b$ , and hypotenuse of  $c$ , then  $a^2 + b^2 = c^2$

**Solution:**

Yes, since  $a, b, c$  are properly specified

6. There are 75 people in this room or the sky is purple

**Solution:**

Yes

**Note**

7 is a **compound proposition**

**Definition 2 (Compound Proposition)**

A **compound proposition** is a proposition that contains two or more **atomic** propositions. In the previous example, the two are:

1. There are 75 people in this room
2. The sky is purple

**Definition 3 (Atomic Proposition)**

An **atomic proposition** is a proposition that cannot be broken down further into separate propositions

**Note**

We denote separate propositions as:  $p, q, r$

### 0.1.2 Logical Operations (connectives)

| Name               | Notation     | Phrasing               | Truth Value                                     |
|--------------------|--------------|------------------------|-------------------------------------------------|
| Disjunction (or)   | $p \vee q$   | "p or q"               | True when $p, q$ or both are <b>TRUE</b>        |
| Conjunction (and)  | $p \wedge q$ | "p and q"              | True when both are <b>TRUE</b>                  |
| Negation (not)     | $\neg p$     | "not p"                | The opposite truth value of $p$                 |
| Exclusive Or (XOR) | $p \oplus q$ | "p or q, but not both" | <b>TRUE</b> when <b>only one</b> is <b>TRUE</b> |

Here are the truth tables of each operation

| $p$      | $q$      | $p \vee q$ | $p \wedge q$ | $\neg p$ | $p \oplus q$ |
|----------|----------|------------|--------------|----------|--------------|
| <b>T</b> | <b>T</b> | <b>T</b>   | <b>T</b>     | <b>F</b> | <b>F</b>     |
| <b>T</b> | <b>F</b> | <b>T</b>   | <b>F</b>     | <b>F</b> | <b>T</b>     |
| <b>F</b> | <b>T</b> | <b>T</b>   | <b>F</b>     | <b>T</b> | <b>T</b>     |
| <b>F</b> | <b>F</b> | <b>F</b>   | <b>F</b>     | <b>T</b> | <b>F</b>     |

**Example 2 (Logical Operators Examples)**

Let  $p$  be the proposition "a is odd" and  $q$  be "a is even" where  $a = 3$

$p$ : 3 is odd = **TRUE**

$q$ : 9 is even = FALSE

Write the following Propositions:

1.  $\neg p$  (3 is even)

**Solution:**  
FALSE

2.  $\neg q$  (9 is odd)

**Solution:**  
TRUE

3.  $p \vee q$  (3 is odd or 9 is even)

**Solution:**  
TRUE since 3 is odd

4.  $p \wedge q$  (3 is odd and 9 is even)

**Solution:**  
FALSE since 9 is not even

5.  $p \wedge \neg q$  (3 is odd and 9 is odd)

**Solution:**  
TRUE since both 3 is odd and 9 is odd

### 0.1.3 Conditionals

| Name          | Notation       | Phrasing                       | Truth Value                              |
|---------------|----------------|--------------------------------|------------------------------------------|
| Conditional   | $p \implies q$ | "p implies q" or "if p then q" | Always True except when $T \implies F$   |
| Biconditional | $p \iff q$     | "p if and only if q"           | True when $p$ and $q$ are the same value |

Here are the truth tables

| $p$ | $q$ | $p \Rightarrow q$ | $p \Leftrightarrow q$ |
|-----|-----|-------------------|-----------------------|
| T   | T   | T                 | T                     |
| T   | F   | F                 | F                     |
| F   | T   | T                 | F                     |
| F   | F   | T                 | T                     |

#### 0.1.4 Logical Transforms

| Name                                | Original          | Output                      |
|-------------------------------------|-------------------|-----------------------------|
| Converse (swap)                     | $p \Rightarrow q$ | $q \Rightarrow p$           |
| Inverse (negate)                    | $p \Rightarrow q$ | $\neg p \Rightarrow \neg q$ |
| Contrapositive (converse + inverse) | $p \Rightarrow q$ | $\neg q \Rightarrow \neg p$ |

Here are the truth tables

| $p$   | $q$   | $p \Rightarrow q$ | $\neg q$ | $\neg p$ | $\neg q \Rightarrow \neg p$ |
|-------|-------|-------------------|----------|----------|-----------------------------|
| TRUE  | TRUE  | TRUE              | FALSE    | FALSE    | TRUE                        |
| TRUE  | FALSE | FALSE             | TRUE     | FALSE    | FALSE                       |
| FALSE | TRUE  | TRUE              | FALSE    | TRUE     | TRUE                        |
| FALSE | FALSE | TRUE              | TRUE     | TRUE     | TRUE                        |

#### Proposition 1 (Contrapositive Correlation)

From the truth table, we see that the converse is logically equivalent to the original expression, so

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$$

#### Definition 4 (Logical Equivalence)

Two propositions are **logically equivalent** if they have the same truth tables

#### Example 3 (Logical Transforms Examples)

Let  $p$  be the proposition "it's cold outside" and  $q$  be "it's snowing outside"

1.  $p \Rightarrow q$  (if it's cold outside, then it's snowing outside)

**Solution:**

FALSE

2.  $q \Rightarrow q$  (converse) (if it's snowing outside, then it's cold outside)

**Solution:**

TRUE

3.  $\neg p \implies \neg q$  (inverse) (if it's not cold outside, then it's not snowing)

**Solution:**

TRUE

4.  $\neg q \implies \neg p$  (contrapositive) (if it's not snowing outside, then it's not cold outside)

**Solution:**

FALSE

**Example 4 (More Examples with Logical Transforms)**

Let  $p$  be the proposition "The triangle is equilateral" and  $q$  be "The triangle is equiangular"

1.  $p \implies q$  (if the triangle is equilateral, then triangle is equiangular")

**Solution:**

TRUE

2.  $q \implies p$  (converse) (if the triangle is equiangular, then the triangle is equilateral)

**Solution:**

TRUE

3.  $\neg p \implies \neg q$  (inverse) (if the triangle is not equiangular, then the triangle is not equilateral)

**Solution:**

TRUE

4.  $\neg q \implies \neg p$  (contrapositive) (if the triangle is not equiangular then the triangle is not equilateral)

**Solution:**

TRUE

**Example 5 (Truth Table Examples)**

1.  $\neg p \vee q$

| $p$ | $q$ | $\neg p$ | $\neg p \vee q$ |
|-----|-----|----------|-----------------|
| T   | T   | F        | T               |
| T   | F   | F        | F               |
| F   | T   | T        | T               |
| F   | F   | T        | T               |

2.  $(p \implies r) \wedge (q \implies r)$

im too lazy to fill this out